BUILDING R.O.B. - VOLUME 5

AI, ROBOTICS, AND CODEX

The accelerated evolution of ROBController and Cerebro

A SOURCE-BASED FIELD GUIDE FOR ADVANCED MAKERS

Rodolfo Aramayo / Orbitus Robotics - First Maker Faire Edition

A human builder remains responsible

Use AI as an engineering collaborator while preserving evidence, review, and physical safety.

This volume reconstructs the software history of ROBController and Cerebro and turns that history into a practical workflow for working with Codex and OpenAI. It distinguishes repository evidence from interpretation, shows how mature control and perception systems evolved, and keeps authority for physical motion with deterministic software and an accountable operator.

The manuscript reflects the repositories inspected on 10 August 2026. Product behavior and guidance can change; verify current OpenAI guidance against official documentation before publication or training.

First evidence-based draft: 2 August 2026. Build details marked **Builder input needed** are deliberate placeholders for the builder to complete.

AI OUTPUT IS NOT MOTION AUTHORITY

Generated code, plans, and tool calls require human review. Test protocol and control changes with fixtures, simulators, lifted treads, strict limits, a spotter, and an independently verified physical emergency stop before operating powered hardware.

ONE CURRENT ARDUINO, THREE HISTORICAL SKETCHES

The `ROBArduino` archive preserves Base, Torso, and Head sketches from an earlier three-Arduino design. The builder reports that present-day ROB uses **only the Base sketch**. This book therefore treats Base behavior as the current low-level firmware reference and labels Torso and Head behavior as retired historical evidence. Source files cannot prove which binary is flashed today; record the board identity, firmware hash, build environment, and upload date before operation.

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CHAPTER 1

Building R.O.B.

AI, Robotics, and the Codex-Accelerated Evolution of ROBController and Cerebro

A source-based field guide
First edition, August 10, 2026

Preface: one builder, many keyboards

R.O.B. did not begin as an "AI-written robot." It began the normal way: one person, several computers, physical hardware, partial ideas, experiments that worked once, experiments that broke connectors, and commits made before a fragile discovery disappeared.

The Git record uses three author-name spellings—`Orbitus`, `RudyAramayo`, and `Rudy Aramayo`—but the same `orbitus@orbitusrobotics.com` email. That is consistent with one builder committing from differently configured computers. It is not evidence of three developers. Nor can Git prove which lines were suggested by an AI. Commit metadata records the person who accepted and committed a change, not every tool involved in producing it.

Still, the history has a visible hinge. Through September 2025, commits read like a lab notebook: tune a servo, repair a port, checkpoint an animation, avoid sending commands too quickly. Beginning August 1, 2026, the repositories gain coordinated transport protocols, certificate pinning, Keychain-backed pairing, formal message schemas, fixture tests, failure boundaries, and long operator documents. ROBController's first large 2026 change adds roughly 5,345 lines and Cerebro's adds roughly 16,478. That is the point at which the work *looks AI-accelerated*.

This book treats August 1, 2026 as the start of the Codex-accelerated era. That is an interpretation based on scale and engineering form, not an authorship claim. Rudy remains the author and operator: the person choosing the goal, checking the diff, testing the hardware, and assuming responsibility for what the robot does.

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1. The robot is a distributed system

R.O.B. is best understood as two cooperating products.

`ROBController` is the operator edge. It runs on Apple mobile devices, accepts human intent, shows state, relays Watch commands, and acts as the approval console. `Cerebro` is the robot-side macOS application. It owns perception, speech, AI sessions, serial hardware, Python helpers, arm scripts, autonomy, stage shows, and the server side of the control link.

The important flow is:

```
human / Apple Watch
```

```
        |
        v
ROBController -- authenticated intent --> Cerebro -- bounded commands --> hardware
        ^                                   |
        |                                   +--> cameras / lidar / speech / AI
        +---------- status and results ----+
```

This architecture implies a rule that should govern every future change:

> Generative AI may propose intent; deterministic software and an accountable operator decide whether that intent can become physical motion.

Robot code has consequences that ordinary application code does not. A UI bug is inconvenient. A stale nonzero tread command, a wrong joint sign, or an unbounded action queue can injure someone or damage hardware. The mature code therefore does more than add intelligence. It constrains intelligence.

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2. Era one: handmade foundations, 2022–2023

ROBController's 2022 root commit established the controller around AutoNet and ROBONet after earlier ML-model experiments were removed. It mixed Objective-C, Objective-C++, Swift, C, and C++, plus Apple's AurioTouch sample-derived audio machinery. This was a practical choice: keep the old hardware-facing code and add modern Apple code where it helped.

In September 2023, the controller gained a custom iPhone storyboard, text commands, lidar visualization, speech-recognition work, system-volume tests, TCP no-delay tuning, layout cleanup, and an early Watch app. These changes are recognizably exploratory. UI, network behavior, and speech lifecycle were being discovered together.

What Codex should learn from this era

Do not begin a legacy-robot task by rewriting the language mix. First ask Codex to map ownership and data flow:

```
Inspect ROBController without changing files. Trace a command from the iPhone
or Watch UI through AutoNet serialization to the network send. Identify the
Objective-C/Swift boundary, thread or queue transitions, and every place a
stale command can survive. Cite files and line numbers. Mark uncertainty.
```

The useful output is a map, not code. Once the map is reviewed, request the smallest change that proves the model correct. This protects working legacy behavior while Codex builds context.

---

3. Era two: the solo workshop, August–September 2025

Cerebro's fresh repository begins August 5, 2025, but its initial import is already a substantial robot program. It includes a macOS storyboard, camera management, speech, serial control, AutoNet, lidar, Kinect/RealSense-era components, Leap Motion, RTSP video, task launchers, Core ML assets, and a large body of mixed Objective-C++ and Swift.

The following weeks read like a maker's engineering diary:

- Gemini is connected, speech delays are adjusted, and OAK camera input begins.
- Camera devices become discoverable at runtime.
- A new upper-neck servo is introduced and calibrated.
- RPLidar launch paths and wake-word handling are repaired.
- Output language propagates from controller to robot voice.
- Amber arm v1/v2 scripts are integrated, corrected, and exercised against real L10 and R11 arms.
- Keyframes and dual-arm sequences grow from manual experiments.
- Vision person segmentation, 3D body pose, SceneKit skeletons, and whole-body overlays are added.
- Speech wordiness, "shut up," camera toggling, and head tracking become operator features.

The commit messages are valuable technical evidence. One warns that arm values are "very scary if the values go wrong." Another records a damaged CAN connector. Another checkpoints code before rebuilding a pose view on every camera frame. These are not embarrassing details. They express the central truth of robotics: hardware provides the final test, and the cost of a bad assumption can be physical.

Turning workshop knowledge into specifications

Before asking Codex to touch hardware code, translate lived knowledge into invariants:

```
Goal: add a head-tilt command.

Constraints:
- Never emit a value outside the calibrated channel-2 range.
- A missing device must disable only head tilt.
- Manual stop must preempt queued motion.
- Preserve existing torso and camera behavior.
- Add a pure serialization fixture before editing the serial path.
- Do not claim hardware validation; give me an exact on-robot checklist.
```

This prompt is good because it tells Codex what must remain true. It does not dictate an implementation before the repository has been inspected.

---

4. The acceleration hinge: August 1, 2026

The paired August 1 commits transform both applications.

In ROBController, the large change introduces secure v2 transport, the Watch companion path, action and Watch command protocols, approval/status surfaces, autonomy controls, tests, and documentation.

In Cerebro, the corresponding change introduces the server side of the secure control plane, Gemini Robotics Live integration, an autonomy coordinator, camera H.264 transport for Vision Pro, a supervised Python runtime, system dependency management, action/video protocols, extensive tests, and a major documentation set.

This is the maturity jump. The old system primarily moved bytes and trusted its environment. The new system names roles, versions messages, authenticates peers, bounds queues, records operator intent, and fails closed.

Why this resembles AI-assisted engineering

No single clue proves AI use. Together, these clues make acceleration a reasonable inference:

1. Change size increases by an order of magnitude.
2. Both sides of a distributed protocol evolve together.
3. Production code arrives with fixture and integration tests.
4. Threat-model concerns—pinning, replay resistance, downgrade prevention, revocation, role checks—are addressed systematically.
5. Operator documentation explains contracts and incomplete capabilities.
6. Later commits use structured conventional-commit language and detailed implementation summaries.

The right conclusion is not "AI replaced the programmer." It is "the builder acquired a fast engineering collaborator, and the repository began to preserve more of the reasoning that previously lived only in the builder's head."

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5. Rebuilding the secure control plane with Codex

The v2 control plane is the strongest example of how to use Codex on a cross-repository change.

5.1 Start with a contract

Ask for a design before edits:

```
Analyze ROBController and Cerebro together. Design a versioned replacement for
the plaintext robot-control path. Requirements: QUIC with TLS 1.3, exact server
certificate pinning, per-device pairing secrets in Keychain, reciprocal proof
of possession, controller and lidar roles, persistent revocation, no automatic
legacy downgrade, bounded framing, and deterministic fixture tests. Identify
the files on both sides and list migration risks. Do not edit yet.
```

Review the design for four separations:

- **Identity:** which Cerebro instance is this?
- **Pairing:** has this device been enrolled?
- **Authorization:** is it an operator or telemetry publisher?
- **Freshness:** is this proof from the current connection rather than a replay?

Only then authorize implementation in checkpoints.

5.2 Implement the shared wire format first

Message formats should have explicit magic bytes, versions, lengths, limits, and error behavior. Generate the same golden fixtures on client and server. Codex is especially useful here because it can compare two implementations and find drift.

```
Implement only the v2 frame codec and fixtures in both repositories. Do not
open sockets and do not modify UI. Reject unknown mandatory versions, oversized
payloads, truncated frames, and invalid role/action combinations. Run the
fixture tests and show the exact diff.
```

5.3 Add transport without weakening authentication

QUIC/TLS encrypts a connection, but encryption alone does not tell a controller that it reached *its* robot. ROBController pins Cerebro's certificate. Pairing then proves possession of a separate per-device secret, while Cerebro resolves the device role from its own registry.

The August 10 fixes demonstrate why network handshakes need state-machine tests. iOS did not always expose Bonjour TXT metadata, and both peers could wait for the other to speak. The repair adds a padded pairing hello before the server sends its challenge, validates the controller ID, and preserves pinning and reciprocal proof.

A productive debugging prompt is:

```
Diagnose the QUIC pairing stall using both repositories. Draw the client and
server state machines from connection start through authenticated readiness.
Find any state where both peers await input. Preserve certificate pinning,
device-role lookup, and reciprocal proof. Add a regression fixture for the
deadlock before changing production code.
```

Never "fix" a handshake by skipping verification or silently falling back to plaintext. Availability bugs must not become authentication bugs.

---

6. Operator authority, Watch control, and autonomy

ROBController becomes more than a joystick in the mature system. It relays Watch intent, advertises accepted action capabilities, approves proposed robot actions, and starts a bounded autonomy session.

The autonomy model is intentionally a session, not indefinite permission. The `social_roam` profile captures an activation pose, stays within a radius, uses fresh lidar, moves slowly, and ends on manual control, explicit stop, or Cerebro restart. Meanwhile `ROBSerialBox` expires controller snapshots after three missed 5 Hz updates, sends one neutral/braked frame, then stops USB writes so the Arduino deadman can de-energize the treads.

This yields a useful authority hierarchy:

```
emergency / stop intent
        > direct operator control
        > bounded approved autonomy
        > AI proposal
        > idle
```

Ask Codex to encode this hierarchy in tests, not just comments:

```
Add table-driven tests for authority arbitration. Stop must preempt every
state. Manual control must end autonomy. Restart must default to autonomy off.
An expired controller snapshot must produce exactly one neutral/braked write
and no continuing heartbeat. Do not add arm movement.
```

The final sentence matters. Mature agents are capable of expanding a coherent feature beyond its safe scope. Tell Codex which adjacent capability is explicitly excluded.

7. Vision, depth, and video as bounded pipelines

Cerebro's camera path evolves from selecting webcams to a multi-provider perception system.

The 2026 implementation runs DepthAI in a supervised Python helper and sends synchronized RGB and aligned depth over a local user-only Unix socket. Keeping the SDK outside Cerebro isolates USB disconnects, malformed packets, missing packages, and Python crashes. AVFoundation remains an RGB fallback, but only one provider may own the OAK device.

The separate Vision Pro service uses its own authenticated QUIC connection, H.264 framing, bounded send state, and newest-frame policy. Slow video cannot back up robot control. Dropped frames trigger keyframe recovery rather than an unbounded queue.

The reusable design lesson is **separate by failure domain**:

- control and video use different services and queues;
- Python SDK faults do not terminate Cerebro;
- camera ownership is exclusive;
- perception drops old work instead of accumulating latency;
- an unavailable optional device degrades one capability.

Codex prompt:

```
Trace camera frames from provider to Vision, Gemini, and Vision Pro. For each
consumer record queue, ownership, backpressure, timestamp, and failure policy.
Then add the smallest change that keeps only the newest pending frame for the
slow consumer. Prove robot-control queues are untouched. Add malformed IPC and
slow-consumer tests; separate simulator evidence from hardware validation.
```

8. AI inside Cerebro: proposals, tools, and speech

The repositories currently document Gemini Robotics Live as the runtime model integration. OpenAI enters this story in two different ways:

1. **Codex develops and maintains the codebase.**
2. **The OpenAI Responses API can power a future model-neutral robot brain.**

These roles should not be conflated. A coding agent may edit a protocol while an application model later uses that protocol as a tool.

For an OpenAI-backed adapter, preserve Cerebro's deterministic boundaries. The model receives observations and returns structured proposals. Cerebro validates the schema, allow-list, session, deadline, and current authority. ROBController approves where required. Only a deterministic executor may touch hardware.

Conceptual Swift-like pseudocode:

```
let proposal = try await openAI.responses.create(
    model: configuredModel,
    input: currentRobotContext,
    tools: [robotActionProposalSchema]
)

let action = try RobotActionProtocol.validate(proposal)
try authority.requireActiveSession(action.sessionID)
try policy.requireAllowed(action)
await operatorConsole.requestApproval(action)
```

Use the Responses API for multi-turn reasoning and tool calling, and choose the model through configuration rather than scattering a model name through UI and network code. Keep credentials outside source control. Do not send secrets, pairing material, faces, audio, or private surroundings unless the product's data policy and operator consent explicitly allow it.

A safe OpenAI integration task for Codex

```
Add an OpenAI Responses API provider behind the existing model-neutral
improvisation/proposal interface. First inspect the current official OpenAI
documentation. Keep the existing provider as default. Read the API key from a
secure runtime source, never source or logs. Return only the existing strict
schema. No model output may directly invoke serial, shell, network-control, or
arm code. Add mocked success, timeout, malformed output, refusal, and
cancellation tests. Do not enable the provider automatically.
```

The official OpenAI model guidance recommends the Responses API for reasoning, tool use, and multi-turn workflows. Models and capabilities change; let Codex consult current official documentation during implementation instead of freezing this book's August 2026 model names into the architecture.

---

9. Local improvisation and stage shows

The August 3 Cerebro change adds a safe local-improvisation layer, a connection-tolerant show coordinator, diagnostics, a llama.cpp provider, strict show schemas, fixture tests, and a Maker Faire opening show.

This feature demonstrates layered trust:

- authored dialogue is the mandatory fallback;
- local generation may select only allow-listed beats and delivery;
- Gemini may add live context to a trusted brief;
- show files cannot contain raw joints, servo values, hosts, ports, or shell;
- uncalibrated gestures fail closed;
- expired stage contexts permanently reject non-stop tool calls;
- `stop_motion` remains available through a priority stop lane.

Generative language becomes safe not because the model is guaranteed to behave, but because untrusted output has a narrow grammar and a deterministic fallback.

When asking Codex to extend a show format, request the validator and negative fixtures before the runner:

```
Extend robshow v1 with an optional audience-response beat. The JSON may select
only a named response policy and timeout. It may not contain executable code,
raw motor values, file paths, hosts, ports, or arbitrary tool names. Update the
schema, parser, negative fixtures, dry-run renderer, and docs before enabling
runtime behavior. Preserve authored fallback and stop preemption.
```

---

10. Dependencies are part of robot safety

Earlier Cerebro code contained machine-specific paths and assumed tools were installed. The mature version treats dependencies as fallible inputs.

Python can be selected, auto-detected, or created in an app-managed virtual environment. Package installation occurs only after an operator action. `sshpass`, `ticcmd`, and RPLidar are resolved and validated before launch. Missing optional tools disable their feature rather than crash Cerebro. Privileged MacPorts installation is shown to the operator in Terminal; the app does not capture an administrator password. The SSH password travels through an anonymous pipe rather than the process argument list.

This is an excellent Codex refactoring pattern:

```
Inventory every Process/NSTask launch in Cerebro. Do not edit yet. Classify the
executable as required or optional; list path source, validation, credentials,
timeout, restart behavior, and user-visible failure. Then centralize only the
shared preflight while preserving each feature's behavior. Add tests for
missing, directory, non-executable, and disappearing executables.
```

Codex can search every launch site more reliably than a tired human, but the operator must decide which dependency failures are tolerable.

---

11. A repeatable Codex workflow for these repositories

Step 1: protect the worktree

Start every session with:

```
Inspect git status in ROBController and Cerebro. Existing changes belong to me.
Do not overwrite, revert, stage, or commit them. Identify generated Xcode user
state separately. Summarize the relevant architecture before editing.
```

The current worktrees, for example, contain modified `UserInterfaceState.xcuserstate` files. They are user state, not permission to discard anything.

Step 2: state outcome and invariants

Prefer "When control packets stop, the base brakes once and becomes silent" to "refactor ROBSerialBox." The former is testable and safety-relevant.

Step 3: ask for read-only diagnosis

For cross-language and cross-repository behavior, have Codex trace the entire path before implementation. Ask it to cite evidence and say what it cannot know without hardware.

Step 4: divide at stable boundaries

Good checkpoints are codec, fixtures, client state machine, server state machine, UI, and documentation. Avoid one request that changes wire format, physical actuation, and user experience without intermediate review.

Step 5: make Codex edit and verify

```
Implement the approved codec checkpoint. Preserve unrelated changes. Run the
narrow fixture tests first, then the relevant build. Review your own diff for
security, concurrency, compatibility, and accidental secrets. Report commands,
results, remaining hardware checks, and files changed.
```

Step 6: inspect the diff yourself

Use `git diff --check`, `git diff --stat`, and `git diff`. Pay special attention to entitlements, Info.plist permissions, Xcode project membership, networking, Keychain accessibility, dispatch queues, and motor values.

Step 7: validate in layers

1. Pure codec/schema fixtures
2. Unit tests with mocked time and I/O
3. Client/server integration tests on loopback
4. Xcode build and static diagnostics
5. Simulator/UI checks where meaningful
6. Bench hardware with motors disabled or lifted
7. Low-speed supervised robot test with an accessible stop

Codex may complete layers one through five. It must never imply that they prove layers six and seven.

Step 8: commit the reasoning

A mature message explains behavior and risk:

```
fix(network): prevent QUIC pairing handshake deadlock

- send a bounded client hello before awaiting the challenge
- validate device identity against the paired registry
- preserve certificate pinning and reciprocal proof
- add a regression fixture for simultaneous-read startup
```

The history then becomes usable context for the next human or AI session.

---

12. Prompt recipes for the actual change families

Cross-repository protocol change

```
Trace protocol X in both repos. Produce a compatibility matrix for old client,
new client, old server, and new server. Design golden fixtures before edits.
Reject malformed and oversized input. No automatic security downgrade. Make
changes in reviewable checkpoints and run both test suites.
```

Concurrency bug

```
Build a state machine from code, queues, callbacks, and cancellation paths.
Identify races, double completion, and mutual waits. Add a deterministic
regression test. Keep UI updates on the main actor/queue and network state on
its owning queue. Do not mask the bug with sleeps.
```

Camera crash

```
Diagnose only first. Trace device discovery, selected device lifetime, session
mutation, and frame consumers. Explain the exact crash precondition. Separate
what can be fixture-tested from the rapid-toggle test required on the robot Mac.
```

New AI tool

```
Treat model output as untrusted. Define a versioned strict schema, limits,
allow-list, authority check, deadline, cancellation, idempotency rule, operator
approval, terminal result, and audit event. Add negative fixtures. Do not wire
physical execution until the protocol review is complete.
```

Arm feature

```
No motion yet. Inventory joint names, signs, units, calibrated ranges, feedback,
URDF assumptions, transforms, collision checks, and emergency stop path. Mark
every missing fact. Build a dry-run command preview and fixtures. Require a
separate explicit request before enabling hardware output.
```

Documentation from code

```
Update the operator document from the implemented diff. Distinguish implemented,
tested in fixtures, built, simulator-tested, and hardware-validated. List known
gaps plainly. Do not describe planned behavior as present behavior.
```

---

13. Reviewing AI-generated robot code

An AI-generated diff is a proposal. Review it in five passes.

Safety

- Does loss of input converge to neutral?
- Is stop independent, prioritized, and idempotent?

- Are speeds, positions, payload sizes, and durations bounded?
- Does restart return to a safe state?
- Does an unavailable sensor prevent motion that depends on it?

Security

- Are identity, enrollment, authorization, and freshness distinct?
- Are secrets absent from source, logs, process arguments, and Bonjour?
- Is there any fallback to plaintext or unauthenticated behavior?
- Can revocation terminate a live session?
- Can a telemetry role become a control role by editing its payload?

Concurrency

- Which queue owns each mutable state variable?
- Can both peers wait forever?
- Can cancellation race with completion or a late tool call?
- Can a slow camera/model consumer block control?

Compatibility

- Are protocol versions explicit?
- Are unknown fields tolerated only where intended?
- Do both repositories share golden fixtures?
- Is legacy behavior opt-in, observable, and removable?

Evidence

- What tests ran?
- What could not run in the environment?
- Was hardware actually exercised?
- Are documentation claims no stronger than the evidence?

---

14. What remains intentionally unfinished

Mature engineering is visible in the features a system refuses to fake. Cerebro reports general picking and arm motion as unavailable because the repository still lacks calibrated camera-to-arm transforms, complete neck mapping, joint feedback, inverse kinematics, collision checking, and a verified grasp executor. Gesture cues fail closed without a calibrated catalog and feedback-capable executor. Approval records operator intent but does not by itself start physical action.

These are the next safe milestones:

1. Establish measured coordinate frames and calibration artifacts.
2. Add feedback and freshness to joint state.
3. Build offline IK with joint and velocity bounds.
4. Add collision models and swept-volume checks.
5. Simulate and dry-run named motions.
6. Validate one arm on a bench at low speed.
7. Introduce a deterministic executor with stop preemption.
8. Only then expose a narrow, schema-constrained AI proposal tool.

Codex can help at every milestone, but it cannot supply missing physical facts from code. Calibration measurements, payload limits, wiring, clearances, and the location of humans around the robot belong to the real world.

---

15. The developer after code generation

Programming is not disappearing so much as moving upward. A developer once spent most of a day translating a known solution into syntax. An AI coding agent can now perform much of that translation, search a repository, connect call sites, write tests, run tools, and revise a patch. The scarce skill becomes the ability to define the right system, communicate its invariants, recognize false confidence, and decide whether the evidence is strong enough to ship.

This is a profound change. Source code remains real and must still execute, but manually typing every line is no longer the center of the job. The new developer operates at several levels at once:

- **Purpose:** What human outcome should improve?
- **System:** Which components, data, authorities, and failure boundaries exist?
- **Contract:** What must inputs, outputs, timing, and errors mean?
- **Evidence:** What observation would prove the behavior works?
- **Operations:** How will the system be deployed, monitored, stopped, and repaired?
- **Responsibility:** Who accepts the consequences when software affects people or machines?

The syntax layer has not become worthless. Reading code is still one way to audit the agent's interpretation, locate a security defect, or understand why a test passes for the wrong reason. But fluency increasingly means being able to move between intent, architecture, executable artifacts, and evidence. A developer can be less concerned with remembering an API name and more concerned with whether the API belongs in the design at all.

From implementation task to outcome contract

A weak request says:

```
Add controller position support.
```

A strong outcome contract says:

```
ROBControllerVision has two spatial controllers. Capture a fresh left and
right world-space pose, preserve chirality, and transmit position in meters
plus normalized quaternion orientation through the authenticated control path.

Cerebro must validate finite values, bounds, version, and freshness before
storing a pose. Missing tracking must invalidate the pose instead of retaining
the last arm target. Preserve the historical tread snapshot for older peers.
Do not actuate an arm yet. Build both apps, add a wire-format fixture, and state
what still requires a physical Vision Pro test.
```

The second request does not prescribe every class or function. It gives the AI room to inspect and design while making the result falsifiable. It names the world in which the code must be correct.

The new literacy

The developer of this wave needs four complementary literacies.

1. **Domain literacy** identifies what matters in robotics, medicine, finance, media, education, or another field.
2. **Systems literacy** traces state, timing, trust, feedback, and failure across boundaries.
3. **AI communication literacy** supplies goals, context, constraints, and acceptance evidence without burying the task in ceremony.
4. **Verification literacy** distinguishes plausible output from demonstrated behavior.

AI magnifies all four. It also magnifies their absence. A vague request can produce a large, polished, internally consistent mistake faster than a person could have typed it.

---

16. How an AI development system actually works

An AI coding system is more than a chat box. It is a loop that combines a model with context, instructions, tools, an execution environment, and feedback. Understanding those parts helps a developer diagnose failures without treating the model as magic.

Model

The model predicts and reasons over the information available in the current interaction. It can synthesize patterns across languages and domains, but it does not automatically know the current repository, physical robot, private requirements, or latest external facts. Confidence in its prose is not a measurement of truth.

Context

Context includes the request, conversation, selected files, repository instructions, tool results, images, logs, and sometimes retrieved documents. Good context is not the largest possible pile of text. It is the smallest set that changes the decision.

For a deadlock, the relevant context is usually both peers' state machines, the first reads and writes, cancellation behavior, and a reproducible trace. Hundreds of unrelated UI files can make the important relationship harder to see. Ask the agent to search first, then load the narrow data path.

Instructions and precedence

An agent may receive durable project instructions in addition to the current request. These can define build commands, architectural rules, formatting, security restrictions, or files that must not be changed. Treat these files as part of the engineering system. Keep them concise, current, and testable.

Instructions should express durable policy:

```
- Run protocol fixture tests after changing either wire implementation.
- Never enable an unauthenticated fallback.
- Do not send arm commands from a generative model directly.
- Preserve user changes in a dirty worktree.
```

Temporary details belong in the task, not the permanent policy file.

Tools

Tools let the agent observe and change the world: search files, inspect Git, apply patches, compile, run tests, browse official documentation, render a PDF, or view an image. A model answer without tools is a proposal. A tool-using agent can produce evidence, although that evidence still needs interpretation.

Tool authority should match the task. Reading and testing are lower risk than publishing, deleting, deploying, spending money, messaging another person, or energizing hardware. High-impact actions need explicit boundaries and often a human approval step.

Execution environment

Local and cloud agents do not necessarily see the same files, credentials, devices, network, simulators, or operating-system APIs. State the important environment facts and ask the agent to report what it could not reproduce. "Build passed" may mean a simulator target compiled; it does not mean two real controllers tracked correctly on Vision Pro.

Feedback loop

The most productive unit of AI development is not one perfect prompt. It is a controlled loop:

```
state goal -> inspect -> form hypothesis -> change -> verify -> review -> revise
```

Each loop should reduce uncertainty. If an agent makes five speculative edits without producing new evidence, stop and return to inspection.

---

17. Communicating for reliable results

Official OpenAI prompting guidance describes four useful ingredients for important work: goal, context, output, and boundaries. It also recommends starting with the result, adding only context that can change it, using follow-up messages, and reviewing the final result yourself. These ideas map especially well to software development.

Goal: define observable behavior

"Improve networking" is a theme. "Reconnect after Wi-Fi interruption without reviving an expired motion lease" is a goal. Use nouns and verbs from the real system. Say which user, device, event, input, and output matter.

When the goal contains multiple interpretations, name a concrete scenario:

```
With Cerebro already listening, launch the Vision client. Both peers currently
wait for the other to speak and the connection times out. Change the handshake
so the client sends a bounded hello first and the server can associate it with
the paired identity before issuing its challenge.
```

Context: provide decision-changing facts

Context is not a biography of the project. Useful context includes:

- reproduction steps and exact error text;
- affected repositories and platforms;
- a known-good comparison implementation;
- protocol fixtures, schemas, or hardware limits;
- recent changes that might explain a regression;
- files that contain user work and must be preserved.

If you believe two bugs share a cause, offer that as a hypothesis rather than a command: "This resembles the QUICK simultaneous-read deadlock we fixed in ROBController; compare the state machines and confirm before reusing the fix." That invites transfer without forcing a false analogy.

Output: say what done looks like

Name the artifacts required at handoff: implementation, migration, tests, documentation, screenshot, fixture, benchmark, or printable PDF. Specify the audience and level when it affects the result. Ask for a concise final report with changed files, test commands, limitations, and the next physical check.

Boundaries: protect what matters

The best boundaries prevent expensive mistakes. They do not micromanage every keystroke.

```
- Keep the legacy 14-line controller snapshot byte-compatible.
- Do not change pairing identity or certificate validation.
- Never retain a stale pose as a valid target.
- Do not actuate hardware in this task.
- Do not discard unrelated working-tree changes.
```

Boundaries can also define when the agent must ask. A request to draft release notes does not authorize publishing them. A request to diagnose a robot motion fault does not authorize sending a test command.

Acceptance evidence

Acceptance criteria turn communication into an engineering contract:

```
Acceptance:
- a valid left/right pose round-trips through the authenticated archive;
- malformed, nonfinite, out-of-range, and unsupported-version poses fail closed;
- an older payload still parses;
- ROBControllerVision and Cerebro Debug builds pass;
- physical-device validation remains clearly marked pending.
```

This is stronger than "make sure it works." It tells both human and AI which claims the evidence must support.

Steering without restarting

When the agent is already working, correct direction with the smallest useful message. Examples:

```
Preserve the existing wire keys; add versioned optional keys instead.
```

```
The green connection light is accurate. Focus on profile classification and
controller aggregation, not discovery.
```

```
Do not stop at storage. Include pose validity and timestamp fields in the model,
but leave arm actuation for a calibrated follow-up.
```

Good steering adds evidence or a constraint. It does not require restating the whole project.

---

18. Context engineering for large repositories

AI performance depends on what it can see and how the repository exposes its meaning. Context engineering is the practice of arranging code, tests, documentation, and instructions so the agent can recover the right model of the system.

Give the repository a map

A short project map should name entry points and ownership:

```
ROBControllerVision/App          UI and session presentation
ROBControlCore/Control           typed intent and dead-man policy
ROBCerebroTransport/Control      authenticated wire compatibility
Cerebro/ROBMainViewController    legacy receive integration
Cerebro/ROBBaseControllerModel   shared accepted controller state
```

This does not replace search. It reduces the chance that the agent edits a similarly named but inactive path.

Preserve executable knowledge

Tests and fixtures are better context than prose alone because they can reject a misunderstanding. For every important promise, ask whether it can become:

- a unit test for a pure transformation;
- a golden byte fixture for a protocol;
- an integration test across a fake transport;
- a state-machine test for cancellation and timeout;
- a simulator scenario for operator behavior;
- a physical checklist for facts software cannot prove.

When a hardware discovery is made, first write it down, then encode the portion that can be checked automatically. “The arm sign is reversed” should become a calibration artifact or mapping test, not remain only in a chat transcript.

Separate durable truth from session history

Conversation is useful working memory but a poor sole archive. After a task, place durable knowledge where the next developer and agent will find it:

- protocol contracts beside the implementation;
- operational steps in a runbook;
- repository conventions in project instructions;
- architectural decisions in a short decision record;
- verified limits in machine-readable configuration;
- uncertain physical facts in a clearly labeled validation checklist.

Avoid copying an entire conversation into the repository. Preserve decisions, evidence, and unresolved questions.

Ask for evidence-backed exploration

For an unfamiliar system, use a staged request:

```
Inspect without editing. Trace a controller sample from GameController through
the dead-man lease, transport encoder, Cerebro receiver, and accepted model.
Cite symbols and files. Identify where chirality, freshness, and invalidation
could be lost. Separate confirmed behavior from hypotheses.
```

Review that map before implementation. This is not bureaucracy; it is a cheap way to discover that the visible UI and active data path are different.

Manage context decay

Long tasks accumulate superseded hypotheses. Periodically restate the current facts:

```
Current verified state:
- connection succeeds and the green indicator is correct;
- both PSVR controllers enumerate;
- the remaining defect is unsupported profile mapping;
- tread values must remain independent floats;
- dead-man behavior is retained.
```

This compact checkpoint helps the agent stop pursuing an earlier diagnosis.

---

19. Verification is the product

AI makes producing an implementation inexpensive. Therefore the value moves to proof. A professional AI-assisted change should carry an evidence ladder whose top rung matches the risk of the claim.

The evidence ladder

1. **Static inspection:** types, call sites, schemas, and configuration agree.
2. **Focused test:** the changed transformation or failure case is executable.

3. **Regression suite:** neighboring promises still hold.
4. **Build:** the real target, SDK, and language mode compile.
5. **Integration:** both sides communicate under representative timing.
6. **Simulation:** user interaction and state transitions behave together.
7. **Bench test:** hardware is constrained, low energy, and supervised.
8. **Operational test:** the complete system runs with stop paths and observers.

Do not skip from static inspection to a claim about physical behavior. Each rung answers a different question.

Ask the agent to falsify its own patch

After implementation, change roles. Ask:

```
Review this diff as a skeptical maintainer. Find ways the new pose can become
stale, swap chirality, bypass validation, break an older Cerebro receiver, or
increase motion authority. Add focused tests for credible failures. Do not
expand scope into arm control.
```

An AI can generate both the patch and the critique, but independence is not guaranteed. The critique still improves coverage by forcing a different search objective. Human review, separate test design, and real measurements remain valuable.

Evaluate repeated AI workflows

When the same task occurs often—triaging controller logs, drafting protocol tests, reviewing dependencies, or generating release notes—build a small eval set. Include normal cases, difficult edge cases, and cases that should be refused or escalated.

Record measurable outcomes:

- Did the agent identify the active implementation?
- Did it preserve backward compatibility?
- Did it distinguish compilation from hardware validation?
- Did it avoid secrets and destructive actions?
- Did the test fail before the fix and pass after it?
- Did the final report accurately describe limitations?

Run the eval when instructions, tools, models, or repository architecture change. A beautiful demonstration is one sample; an eval set measures a pattern.

Stop conditions are part of verification

Define when work must pause:

- the requested behavior depends on an unknown physical limit;
- the only available test could move unguarded hardware;
- credentials or permissions are missing;

- a generated migration could irreversibly alter user data;
- two authoritative specifications conflict;
- the diff overlaps unexplained user changes.

Stopping with a precise blocker is a successful safety behavior, not a failure of intelligence.

---

20. Designing AI-native software systems

Using AI to write an ordinary application is only the first wave. An AI-native system is designed around uncertain model output, typed tools, observable state, evals, and deterministic boundaries.

Separate proposal from execution

The model should propose a bounded intent:

```
turn 20 degrees left at inspection speed
```

A deterministic layer should decide whether that intent is allowed, translate it into units, apply limits, acquire authority, monitor freshness, and stop. The model should not manufacture raw motor bytes or bypass the controller lease.

For an arm, a safer progression is:

```
language goal
  -> typed task proposal
  -> scene and capability checks
  -> deterministic planner / IK
  -> joint, velocity, and collision validation
  -> operator approval when required
  -> feedback-controlled executor
  -> independent stop
```

Each arrow is an inspectable contract.

Give tools narrow schemas

A tool named `run_any_shell_command` offers enormous accidental authority. A robot tool should expose the smallest operation that can be made safe. Its schema should constrain units, ranges, identifiers, and optionality before execution code sees the request.

```
propose_named_pose(
  pose_id: one of the calibrated catalog IDs,
  speed_fraction: 0.0 through 0.2,
  reason: short operator-visible text
)
```

Even a valid schema does not prove the pose is currently collision-free. Tool validation, world state, executor feedback, and operator policy remain separate layers.

Treat retrieval as evidence selection

An AI system can retrieve manuals, logs, code, and prior decisions. Retrieval quality determines which evidence reaches the model. Store documents with source, date, version, authority, and scope. Prefer a

current verified wiring map over an old brainstorming note, and make conflicts visible rather than silently blending them.

Design for uncertainty

Model output may be incomplete, inconsistent, or wrong. The system needs:

- structured outputs that reject invalid shapes;
- timeouts and cancellation;
- bounded retries and queue sizes;
- idempotent operations where possible;
- explicit unavailable and uncertain states;
- audit events for proposals, approvals, actions, and outcomes;
- human escalation for consequential ambiguity.

Do not convert "the model did not answer" into a default motion. In robotics, absence of valid intent should converge to neutral.

Observe the whole loop

Measure more than model latency. Track tool selection, validation failures, approval delays, execution outcomes, cancellation, stale state, and recovery. Keep sensitive data out of logs and define retention. An AI system that cannot explain which source and tool produced an action will be difficult to debug and unsafe to trust.

---

21. A thirty-day AI developer practice

The fastest way to learn this new form of development is to practice on real, bounded work while increasing authority slowly.

Week 1: learn to specify

Each day, take one vague task and write an outcome contract with goal, context, output, boundaries, and acceptance evidence. Ask the AI to inspect before editing. Compare its system map with the code.

Exercises:

1. Trace one UI action to its side effect.
2. Explain one protocol field and its validation.
3. Find one timeout and list what state it protects.
4. Turn one bug report into reproducible steps.
5. Convert one workshop fact into a testable invariant.

Week 2: build evidence loops

Choose low-risk bugs. Require a failing reproduction or fixture before the patch. Ask the agent to run the smallest test, then the neighboring suite, then the real build. Review every claim in the handoff against command output.

Keep a notebook with three columns: claim, evidence, remaining uncertainty. This trains the distinction between "the code looks right" and "this behavior was observed."

Week 3: direct cross-system changes

Work across two components without expanding physical authority. Examples are a versioned optional field, a status message, or a simulator-only control. Require a shared fixture and independent validation at the receiver.

Practice steering when the first hypothesis is wrong. Give the agent the new fact without rewriting its entire plan. Watch whether it updates the causal model or merely patches around the symptom.

Week 4: design an AI-native capability

Design—but do not energize—a narrow AI proposal tool. Define its schema, authorization, freshness, deterministic checks, refusal cases, audit record, and eval set. Threat-model prompt injection, stale retrieval, malformed tool arguments, repeated calls, cancellation races, and operator confusion.

Finish with a review packet:

- system diagram;
- typed contract;
- ten normal eval cases;
- ten adversarial or failure cases;
- deterministic validator tests;
- simulator demonstration;
- physical validation plan owned by a qualified operator.

Graduation test

You are ready for greater autonomy when you can reliably answer:

1. What exact outcome is requested?
2. Which source is authoritative for each fact?
3. What authority does the AI have?
4. What deterministic boundary contains it?
5. What evidence supports each completion claim?
6. How does the system fail neutral?
7. Who can stop it, and how?

The future developer is not the person who types the most code. It is the person who can turn human intent into a well-bounded system, use AI to explore and construct it rapidly, and produce evidence strong enough for others to trust.

---

22. The AI change contract worksheet

Complete this worksheet before a consequential AI-assisted change. Short, specific answers are more valuable than polished language.

Outcome

- Who experiences the problem?
- What behavior should become observably different?
- What example demonstrates success?
- What is explicitly outside this change?

Sources and system map

- Which repository, branch, target, and device are involved?
- Which files or symbols are likely entry points?
- What specification, fixture, measurement, or operator statement is authoritative?
- Which facts are hypotheses that the agent must confirm?
- What recent known-good implementation should be compared?

Invariants and authority

- What existing behavior must remain byte-compatible or user-compatible?
- Which values, rates, sizes, lifetimes, and coordinate systems are bounded?
- Who owns mutable state?
- What may the AI read, edit, execute, or propose?
- Which actions require approval?
- Which actions are forbidden in this task?

Failure behavior

- What happens when input disappears, tracking is lost, or a peer disconnects?
- What happens when a value is malformed, nonfinite, stale, duplicated, or unsupported?
- Can cancellation race with completion?
- Can one optional subsystem block control or stop handling?
- Does restart preserve an emergency stop or accidentally clear it?

Acceptance evidence

- What test must fail before the fix?
- What focused test proves the new contract?
- What regression suites and builds must pass?
- What integration or simulator scenario must be observed?
- What remains a physical-device test?
- What limitation must appear in the final report?

Ready-to-use prompt

```
Goal:
[Describe the observable result.]

Context:
[Name the repositories, reproduction, relevant evidence, and known-good comparison.]

Boundaries:
[List the few invariants and authority limits that prevent real harm.]

Acceptance:
[List executable tests, builds, and remaining physical checks.]

Start by inspecting the active data path and reporting confirmed facts versus
hypotheses. Then implement the smallest complete change, verify it, review the
diff for regressions and unsafe authority expansion, and give me an evidence-
backed handoff. Do not claim validation that the available environment cannot
perform.
```

Human sign-off

Before shipping or energizing hardware, the responsible developer should be able to say: I understand the intended behavior; I reviewed the authority and failure boundaries; the cited evidence supports the claims; unresolved physical facts are labeled; and a tested independent stop exists where motion can affect people or equipment.

Epilogue: authorship after acceleration

The early repositories preserve the voice of one person working from several machines: informal, immediate, hardware-aware, and willing to checkpoint an uncertain experiment. The newer code does not erase that voice. It gives it leverage.

The mature system is better not merely because it contains more code. It has contracts where there were assumptions, bounded queues where there could be backlogs, explicit authority where there was trust, tests where there was only memory, and documentation where knowledge once lived in the workshop.

That is the best use of Codex in robotics. Let AI search widely, trace deeply, draft quickly, compare both sides of a protocol, and remember every edge case you name. Keep the human responsible for scope, evidence, physical validation, and the decision to energize the machine.

Sources and further reading

Repository sources

- `ROBController` Git history through `eb57375`, August 10, 2026
- `Cerebro` Git history through `d45f9c9`, August 10, 2026
- `ROBController/docs/rob-control-v2-and-autonomy.md`
- `ROBController/docs/robot-action-console.md`
- `ROBController/docs/watch-controller.md`
- `Cerebro/docs/rob-control-v2.md`
- `Cerebro/docs/controller-activated-autonomy.md`
- `Cerebro/docs/depth-camera.md`
- `Cerebro/docs/vision-pro-video.md`
- `Cerebro/docs/gemini-robotics-live.md`
- `Cerebro/docs/gemini-robotics-stage-action-plan.md`
- `Cerebro/docs/local-improvisation-provider.md`

Official OpenAI documentation

- Prompting for ChatGPT and Codex[1]
- Codex use cases[2]
- OpenAI model guidance[3]
- OpenAI API models[4]

Consult the current official documentation when implementing; models, capabilities, availability, and SDK shapes can change after this edition.

[1] https://learn.chatgpt.com/docs/prompting
[2] https://developers.openai.com/codex/use-cases
[3] https://developers.openai.com/api/docs/guides/latest-model
[4] https://developers.openai.com/api/docs/models